Hands-On-Activity No. 4.3	
Title **STRUCTURES**	
Course Code: CPE 007	**Program: Computer Engineering**
Course Title: Programming Logic and Design	**Date Performed: October 8, 2026**
Section: 11S2	**Date Submitted: October 8, 2026**
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1. Objective(s)

This activity aims to develop programming skills using structures.

2. Intended Learning Outcomes

This activity aims to develop programming skills using functions.

3. Materials and Equipment

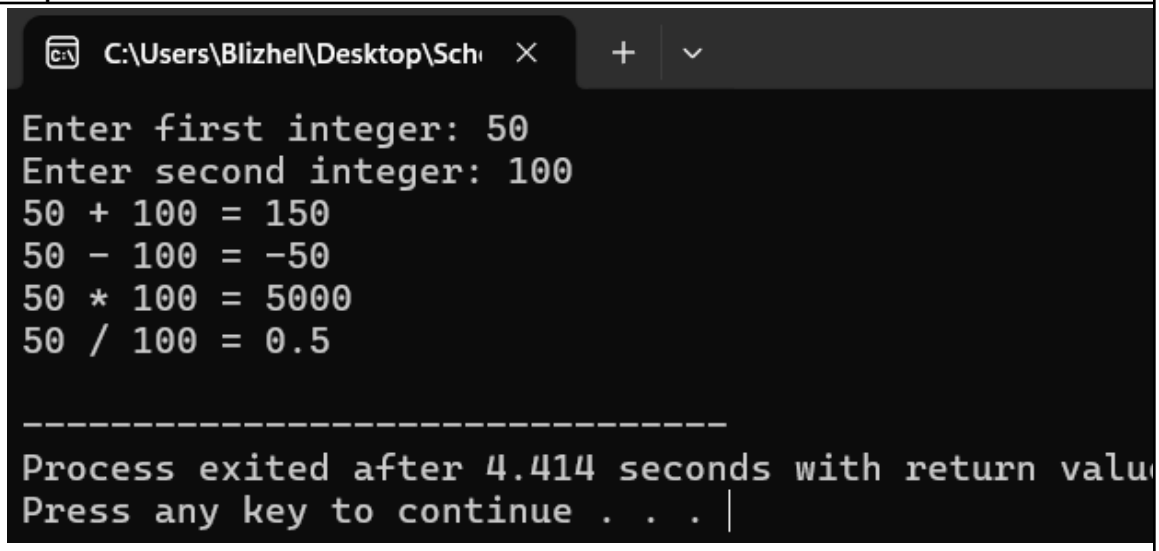
Internet connection
Personal Computer

4. Procedure

1. 2.1 Create a program in C++ that will add, subtract, divide, multiply for a 2 user-input integer values. Use functions in the program.
2. 2.2 Create a program in C++ that will convert a user-input Farenheight to Celcius and vice versa. Use functions in the program.
3. 2.3 Create a program in C++ that will convert dollars into pesos and vice versa. Use functions in the program.

5. Output

1.




```
#include <iostream>
using namespace std;

//returns sum of two integers
int add (int a, int b) {return a + b; }
// returns difference of two integers
int substract (int a, int b) {return a - b; }
// returns product of two integers
int multiply (int a, int b) {return a * b; }
```

```
//returns quotient of two integers as a double
double divide (int a, int b) {
	return (double) a / b;
}

int main ()
{
	int a, b; // variable to store two value entered by user

	cout << "Enter first integer: ";
	cin >> a;
	cout << "Enter second integer: ";
	cin >> b;

	// displays the result
	cout << a << " + " << b << " = " << add (a, b) << endl;
	cout << a << " - " << b << " = " << substract (a, b) << endl;
	cout << a << " * " << b << " = " << multiply (a, b) << endl;

	// only divide if the second number is not zero
	if (b != 0) {
		cout << a << " / " << b << " = " << divide (a, b) << endl;
	} else  {
		cout << "Can not be divided." << endl;

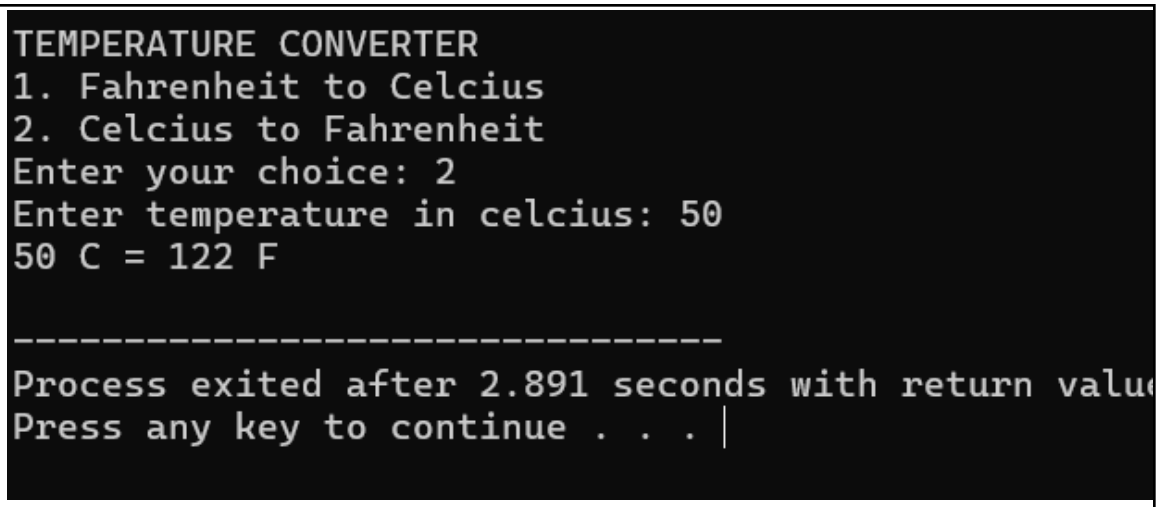
	}

	return 0;
}
```

2.

```
TEMPERATURE CONVERTER
1. Fahrenheit to Celcius
2. Celcius to Fahrenheit
Enter your choice: 1
Enter temperature in fahrenheit: 70
70 F = 21.1111 C

--------------------------------
Process exited after 7.879 seconds with return value
Press any key to continue . . .
```




```
#include <iostream>
using namespace std;

// converts temperature from fahrenheit to celcius
double fahrenheittoCelcius (double f) {
	return (f - 32) * 5.0 / 9.0;
}

// converts temperature from celcius to fahrenheit
double celciustoFahrenheit (double c) {
	return (c * 9.0 / 5.0) + 32;
}

int main () {
	int choice; // stores users choice
	double temp;

	// display menu
	cout << "TEMPERATURE CONVERTER" << endl;
	cout << "1. Fahrenheit to Celcius" << endl;
	cout << "2. Celcius to Fahrenheit" << endl;
	cout << "Enter your choice: ";
	cin >> choice;

	// fahrenheit to celcius
	if (choice == 1) {
		cout << "Enter temperature in fahrenheit: ";
		cin >> temp;
		cout << temp << " F = " << fahrenheittoCelcius (temp) << " C" << endl;
	// celcius to fahrenheit
	} else if (choice == 2) {
		cout <<"Enter temperature in celcius: ";
		cin >> temp;
		cout << temp << " C = " << celciustoFahrenheit (temp) << " F" << endl;
	} else {
		cout << "Invalid choice." << endl;
```

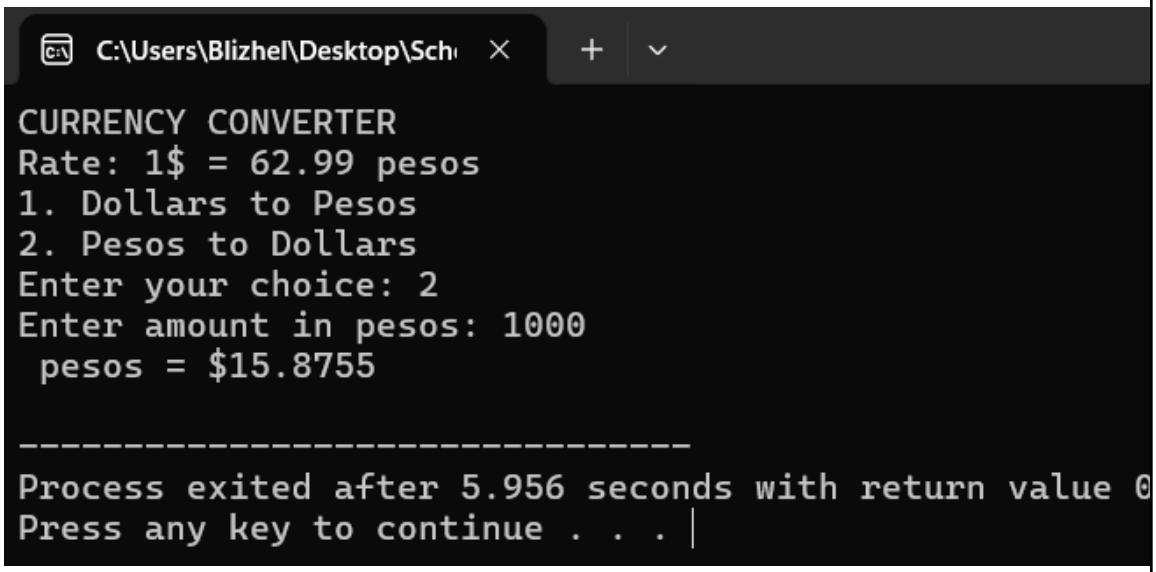
```
        }

        return 0;
}
```

```
CURRENCY CONVERTER
Rate: 1$ = 62.99 pesos
1. Dollars to Pesos
2. Pesos to Dollars
Enter your choice: 1
Enter amount in dollars: 500
$500 = 31495 pesos

--------------------------------
Process exited after 8.262 seconds with return value
Press any key to continue . . .
```

3.




```
#include <iostream>
using namespace std;

const double RATE = 62.99;

double dollarstoPesos (double dollars) {
        return dollars * RATE;
}

double pesostoDollars (double pesos) {
        return pesos / RATE;
}

int main ()
```

```
{
        int choice;
        double amount;

        //display menu
        cout << "CURRENCY CONVERTER" << endl;
        cout << "Rate: 1$ = " << RATE << " pesos" << endl;
        cout << "1. Dollars to Pesos" << endl;
        cout << "2. Pesos to Dollars" << endl;
        cout << "Enter your choice: ";
        cin >> choice;

        if (choice == 1) {
                cout << "Enter amount in dollars: ";
                cin >> amount;
                cout << "$" << amount << " = " << dollarstoPesos (amount) << " pesos" << endl;
        } else if (choice == 2) {
                cout << "Enter amount in pesos: ";
                cin >> amount;
                cout << " pesos = $" << pesostoDollars (amount) << endl;
        } else {
                cout << "Invalid choice.";
        }

        return 0;
}
```

6. Conclusion

In this hands-on activity I learned about how to properly create a program that uses functions.

7. Rubric

Rubric for SO 7 (14)							
Criteria	**Ratings**						**Pts**
SO 7 PI 1 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently and applies knowledge learned into practice	5 pts Good | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently	4 pts Satisfactory | Look beyond classroom requirements, showing interest in pursuing knowledge independently	3 pts Unsatisfactory | Begins to look beyond classroom requirements, showing interest in pursuing knowledge independently	2 pts Poor | Relies on classroom instruction only	1 pts Very Poor | No initiative or interest in acquiring new knowledge	6 pts
SO 7 PI 2 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Completes an assigned task independently and practices continuous improvement	5 pts Good | Completes an assigned task without supervision or guidance	4 pts Satisfactory | Requires minimal guidance to complete an assigned task	3 pts Unsatisfactory | Requires detailed or step-by-step instructions to complete a task	2 pts Poor | Shows little interest to complete a task independently	1 pts Very Poor | No interest to complete a task independently	6 pts
SO 7 PI 3 IILO4 Utilize lifelong learning skills in pursuit of personal development and excellence in professional practice. threshold: 4.8 pts	6 pts Excellent | Synthesizes and integrates information from a variety of sources; formulates a clear and precise perspective; draws appropriate conclusions	5 pts Good | Evaluate information from a variety of sources; formulates a clear and precise perspective.	4 pts Satisfactory | Analyze information from a variety of sources; formulates a clear and precise perspective.	3 pts Unsatisfactory | Apply the gathered information to formulate the problem	2 pts Poor | Gather and summarized the information from a variety of sources but failed to formulate the problem	1 pts Very Poor | Gather information from a variety of sources	6 pts